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In [1]: # Practice Program-7
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# Name: Ashlesha Tayade
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Name:Ruchika Gaikwad
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In [2]: # Topic : Plotly
# 1.Draw a bar chart showing GDP percentage for European countires. [use gdpmindr dataset]
#       Use different semantics as orientation, bar mode, bar norm, text to improve your chart
#       For the same dataset show GDP over the years using animation frame as Country
# 2.Draw a Histogram showing votes secured by 3 candidates - Coderre, Joly, Bergeron
#       Use semantics -bar mode, color, facet_row to improve your chart. Create a cumulative grap
# 3.Draw a scatter plot to show relationship between iris petal length and sepal width
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In [3]: import plotly.express as px
import seaborn as sns
```

```
In [4]: dir(px.data)
```

```
Out[4]: ['AVAILABLE_BACKENDS',
'BACKENDS_WITH_INDEX_SUPPORT',
'__all__',
'__builtins__',
'__cached__',
'__doc__',
'__file__',
'__loader__',
'__name__',
'__package__',
'__path__',
'__spec__',
'carshare',
'election',
'election_geojson',
'experiment',
'gapminder',
'import_module',
'iris',
'medals_long',
'medals_wide',
'nw',
'os',
'stocks',
'tips',
'wind']
```

```
In [5]: # 1.Draw a bar chart showing GDP percentage for European countires. [use gdpmindr dataset]
#       Use different semantics as orientation, bar mode, bar norm, text to improve your chart
#       For the same dataset show GDP over the years using animation frame as Country
df = px.data.gapminder()
df
```

Out[5]:

	country	continent	year	lifeExp	pop	gdpPercap	iso_alpha	iso_num
0	Afghanistan	Asia	1952	28.801	8425333	779.445314	AFG	4
1	Afghanistan	Asia	1957	30.332	9240934	820.853030	AFG	4
2	Afghanistan	Asia	1962	31.997	10267083	853.100710	AFG	4
3	Afghanistan	Asia	1967	34.020	11537966	836.197138	AFG	4
4	Afghanistan	Asia	1972	36.088	13079460	739.981106	AFG	4
...	...	...	...	...	...	...	...	...
1699	Zimbabwe	Africa	1987	62.351	9216418	706.157306	ZWE	716
1700	Zimbabwe	Africa	1992	60.377	10704340	693.420786	ZWE	716
1701	Zimbabwe	Africa	1997	46.809	11404948	792.449960	ZWE	716
1702	Zimbabwe	Africa	2002	39.989	11926563	672.038623	ZWE	716
1703	Zimbabwe	Africa	2007	43.487	12311143	469.709298	ZWE	716

1704 rows × 8 columns

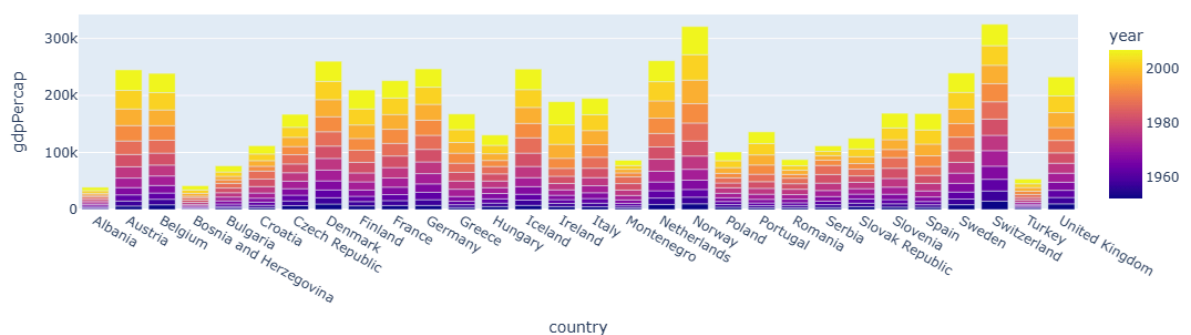
```
In [6]: filt= df.loc[df["continent"]=="Europe"]
        filt
```

Out[6]:

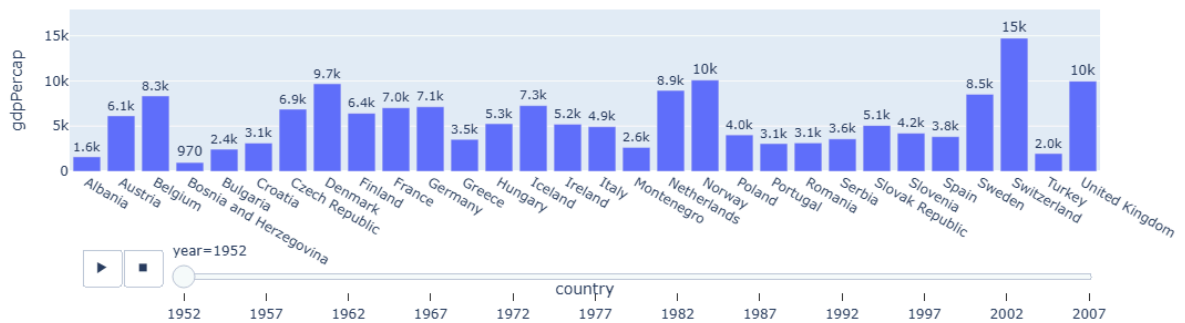
	country	continent	year	lifeExp	pop	gdpPercap	iso_alpha	iso_num
12	Albania	Europe	1952	55.230	1282697	1601.056136	ALB	8
13	Albania	Europe	1957	59.280	1476505	1942.284244	ALB	8
14	Albania	Europe	1962	64.820	1728137	2312.888958	ALB	8
15	Albania	Europe	1967	66.220	1984060	2760.196931	ALB	8
16	Albania	Europe	1972	67.690	2263554	3313.422188	ALB	8
...	...	...	...	...	...	...	...	...
1603	United Kingdom	Europe	1987	75.007	56981620	21664.787670	GBR	826
1604	United Kingdom	Europe	1992	76.420	57866349	22705.092540	GBR	826
1605	United Kingdom	Europe	1997	77.218	58808266	26074.531360	GBR	826
1606	United Kingdom	Europe	2002	78.471	59912431	29478.999190	GBR	826
1607	United Kingdom	Europe	2007	79.425	60776238	33203.261280	GBR	826

360 rows × 8 columns

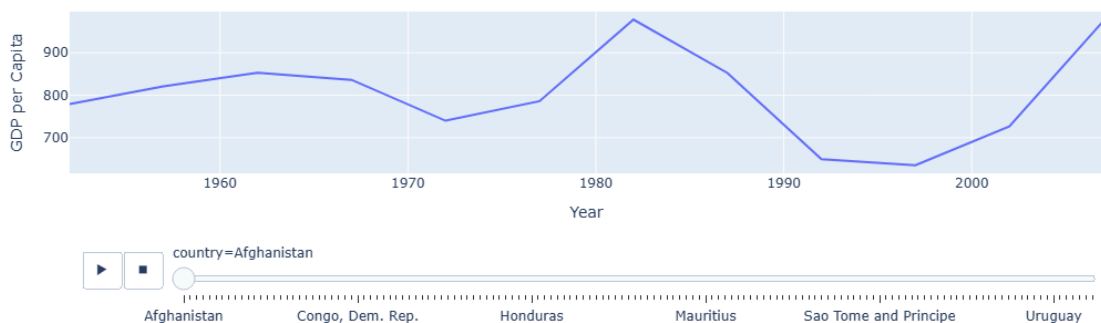
```
In [7]: fig=px.bar(filt,x=filt['country'],y=filt['gdpPercap'],color=filt['year'])
        fig.show()
```



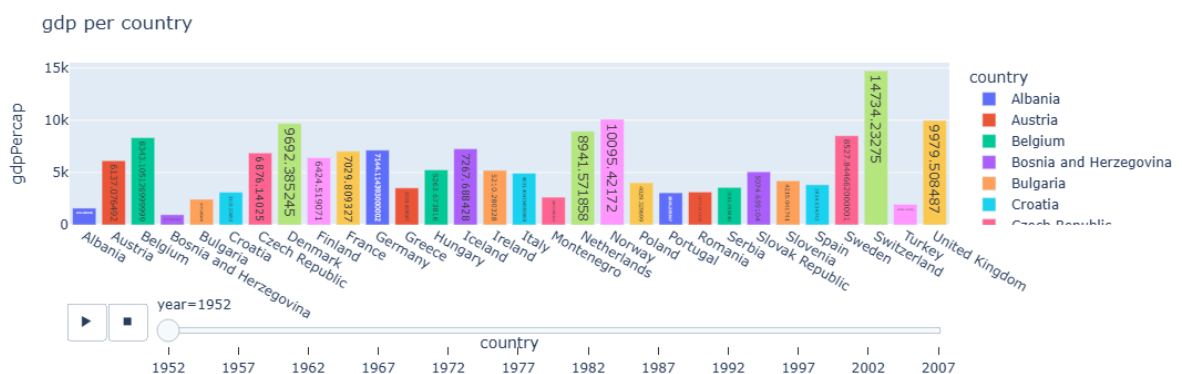
```
In [8]: fig=px.bar(filt,x=filt['country'],y=filt['gdpPercap'],animation_frame=filt['year'],text_auto='.2s')
fig.update_traces(textfont_size=12, textangle=0, textposition="outside", cliponaxis=False)
fig.show()
```



```
In [9]: fig = px.line(df, x=df['year'], y=df["gdpPercap"], animation_frame=df["country"], )
fig.update_layout(xaxis_title="Year", yaxis_title="GDP per Capita")
fig.show()
```



```
In [10]: fig=px.bar(filt, x='country', y='gdpPercap', text='gdpPercap', animation_frame='year', color='country')
fig.update_layout(title='gdp per country',showlegend = True,hovermode='y unified')
fig.show()
```



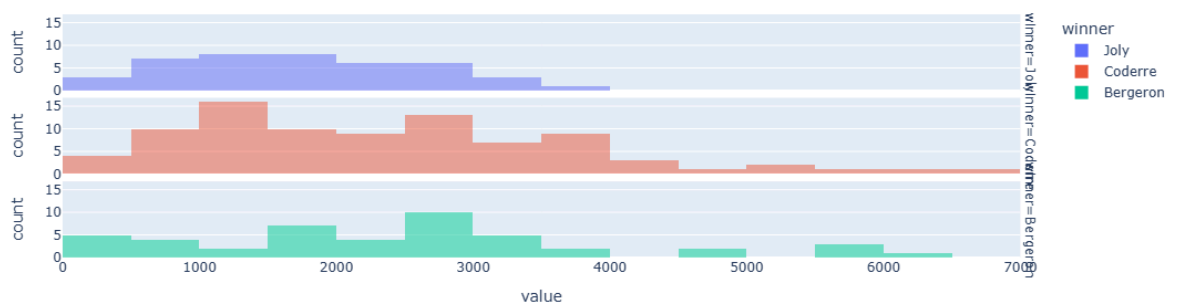
```
In [11]: # 2.Draw a Histogram showing votes secured by 3 candidates - Coderre, Joly, Bergeron
#         Use semantics -bar mode, color, facet_row to improve your chart. Create a cumulative graph
df=px.data.election()
df
```

Out[11]:

	district	Coderre	Bergeron	Joly	total	winner	result	district_id
0	101-Bois-de-Liesse	2481	1829	3024	7334	Joly	plurality	101
1	102-Cap-Saint-Jacques	2525	1163	2675	6363	Joly	plurality	102
2	11-Sault-au-Récollet	3348	2770	2532	8650	Coderre	plurality	11
3	111-Mile-End	1734	4782	2514	9030	Bergeron	majority	111
4	112-DeLorimier	1770	5933	3044	10747	Bergeron	majority	112
5	113-Jeanne-Mance	1455	3599	2316	7370	Bergeron	plurality	113
6	12-Saint-Sulpice	3252	2521	2543	8316	Coderre	plurality	12
7	121-La Pointe-aux-Prairies	5456	1760	3330	10546	Coderre	majority	121
8	122-Pointe-aux-Trembles	4734	1879	2852	9465	Coderre	majority	122
9	123-Rivière-des-Prairies	5737	958	1656	8351	Coderre	majority	123
10	13-Ahuntsic	2979	3430	2873	9282	Bergeron	plurality	13
11	131-Saint-Édouard	1827	6408	2815	11050	Bergeron	majority	131
12	132-Étienne-Desmarteau	2331	5748	2788	10867	Bergeron	majority	132
13	133-Vieux-Rosemont	2670	4962	3234	10866	Bergeron	plurality	133
14	134-Marie-Victorin	3673	3155	2431	9259	Coderre	plurality	134
15	14-Bordeaux-Cartierville	3612	1554	2081	7247	Coderre	plurality	14
16	141-Côte-de-Liesse	4308	1320	3959	9587	Coderre	plurality	141
17	142-Norman-McLaren	4104	1459	2822	8385	Coderre	plurality	142
18	151-Saint-Léonard-Est	3931	882	1641	6454	Coderre	majority	151
19	152-Saint-Léonard-Ouest	5387	1184	1908	8479	Coderre	majority	152
20	161-Saint-HenriPetite-BourgognePointe-Saint-Ch...	2432	3368	3578	9378	Joly	plurality	161
21	162-Saint-PaulÉmard	2566	2092	2438	7096	Coderre	plurality	162
22	171-ChamplainL'Île-des-Soeurs	3347	2562	3291	9200	Coderre	plurality	171
23	172-Desmarchais-Crawford	2476	2631	2849	7956	Joly	plurality	172
24	181-Peter-McGill	1451	754	1894	4099	Joly	plurality	181
25	182-Saint-Jacques	1906	2169	2282	6357	Joly	plurality	182
26	183-Sainte-Marie	1347	2827	2271	6445	Bergeron	plurality	183
27	191-Saint-Michel	3668	984	1220	5872	Coderre	majority	191
28	192-François-Perrault	2878	2666	2039	7583	Coderre	plurality	192
29	193-Villeray	2201	5819	2782	10802	Bergeron	majority	193
30	194-Parc-Extension	2420	1793	1402	5615	Coderre	plurality	194
31	21-Ouest	2184	691	1076	3951	Coderre	majority	21
32	22-Est	1589	708	1172	3469	Coderre	plurality	22
33	23-Centre	2526	851	1286	4663	Coderre	majority	23
34	31-Darlington	1873	1182	1232	4287	Coderre	plurality	31
35	32-Côte-des-Neiges	1644	1950	1578	5172	Bergeron	plurality	32
36	33-Snowdon	1548	1503	1636	4687	Joly	plurality	33
37	34-Notre-Dame-de-Grâce	1773	2653	3262	7688	Joly	plurality	34

	district	Coderre	Bergeron	Joly	total	winner	result	district_id
38	35-Loyola	2040	1437	2648	6125	Joly	plurality	35
39	41-du Canal	1165	832	1266	3263	Joly	plurality	41
40	42-J.-Éméry-Provost	1193	653	1157	3003	Coderre	plurality	42
41	43-Fort-Rolland	1325	1205	1908	4438	Joly	plurality	43
42	51-Sault-Saint-Louis	4201	1642	3717	9560	Coderre	plurality	51
43	52-Cecil-P.-Newman	3536	1330	2943	7809	Coderre	plurality	52
44	61-Pierre-Foretier	631	258	998	1887	Joly	majority	61
45	62-Denis-Benjamin-Viger	595	226	1068	1889	Joly	majority	62
46	63-Jacques-Bizard	518	224	690	1432	Joly	plurality	63
47	64-Sainte-Genève	332	131	326	789	Coderre	plurality	64
48	71-Tétreaultville	3694	2589	3454	9737	Coderre	plurality	71
49	72-MaisonneuveLongue-Pointe	2746	3250	3139	9135	Bergeron	plurality	72
50	73-Hochelaga	1546	3679	2675	7900	Bergeron	plurality	73
51	74-Louis-Riel	3509	2178	2338	8025	Coderre	plurality	74
52	81-Marie-Clara	6591	1085	1435	9111	Coderre	majority	81
53	82-Ovide-Clermont	6229	780	1051	8060	Coderre	majority	82
54	91-Claude-Ryan	996	643	423	2062	Coderre	plurality	91
55	92-Joseph-Beaubien	540	833	592	1965	Bergeron	plurality	92
56	93-Robert-Bourassa	446	465	419	1330	Bergeron	plurality	93
57	94-Jeanne-Sauvé	491	698	489	1678	Bergeron	plurality	94

```
In [12]: fig=px.histogram(df,x=['Coderre', 'Joly', 'Bergeron'],color='winner',barmode='overlay',facet_row='winner')
fig.show()
```



```
In [13]: # 3.Draw a scatter plot to show relationship between iris petal length and sepal width
df = px.data.iris()
df
```

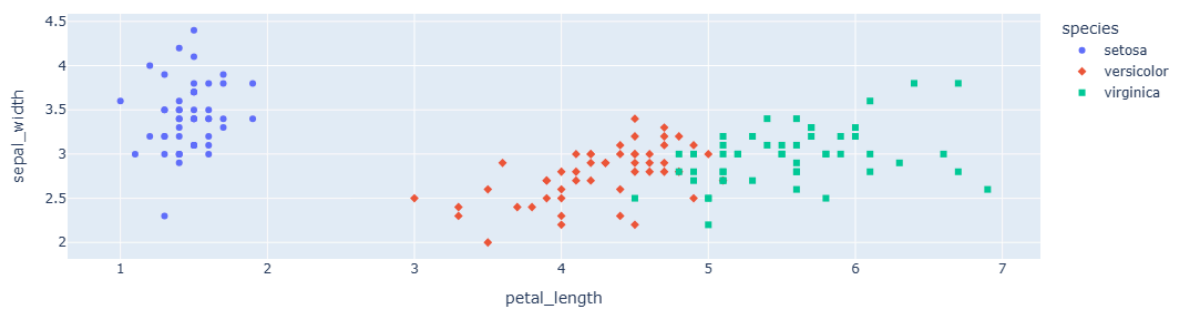
Out[13]:

	sepal_length	sepal_width	petal_length	petal_width	species	species_id
0	5.1	3.5	1.4	0.2	setosa	1
1	4.9	3.0	1.4	0.2	setosa	1
2	4.7	3.2	1.3	0.2	setosa	1
3	4.6	3.1	1.5	0.2	setosa	1
4	5.0	3.6	1.4	0.2	setosa	1
...	...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	virginica	3
146	6.3	2.5	5.0	1.9	virginica	3
147	6.5	3.0	5.2	2.0	virginica	3
148	6.2	3.4	5.4	2.3	virginica	3
149	5.9	3.0	5.1	1.8	virginica	3

150 rows × 6 columns

```
In [14]: fig=px.scatter(df, x='petal_length', y='sepal_width', color='species', symbol='species')
```

```
In [15]: fig.show()
```



In []: